EXAMEN MÓDULO 2

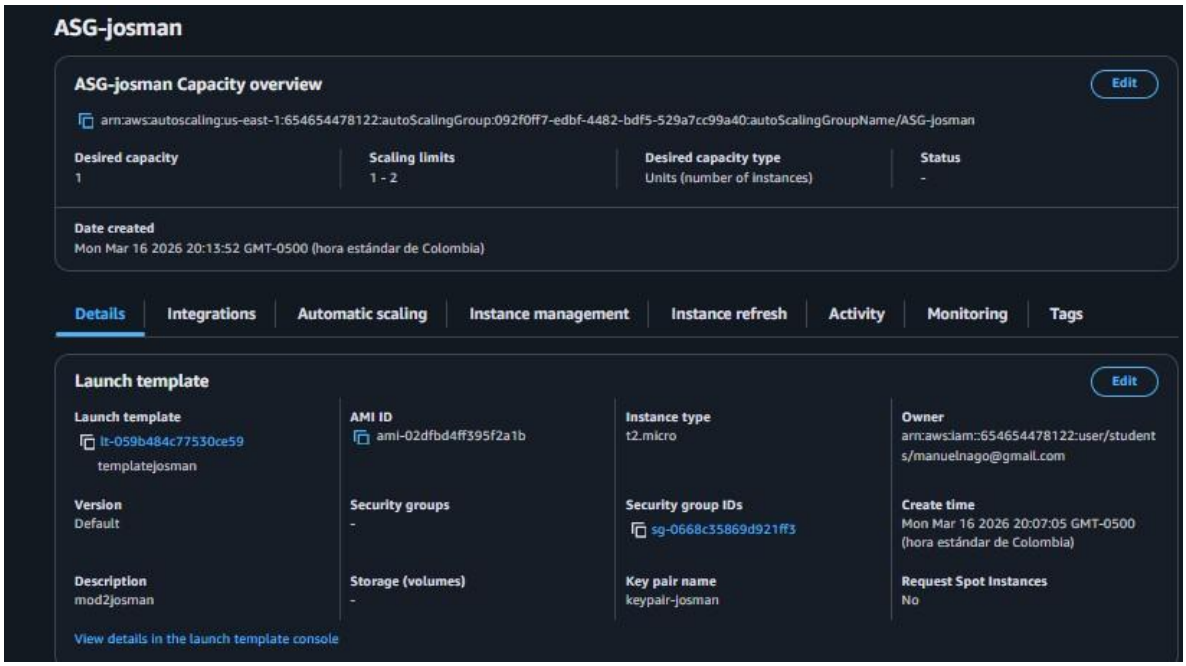
Parte 1: Configurar la Escalabilidad (Auto Scaling)

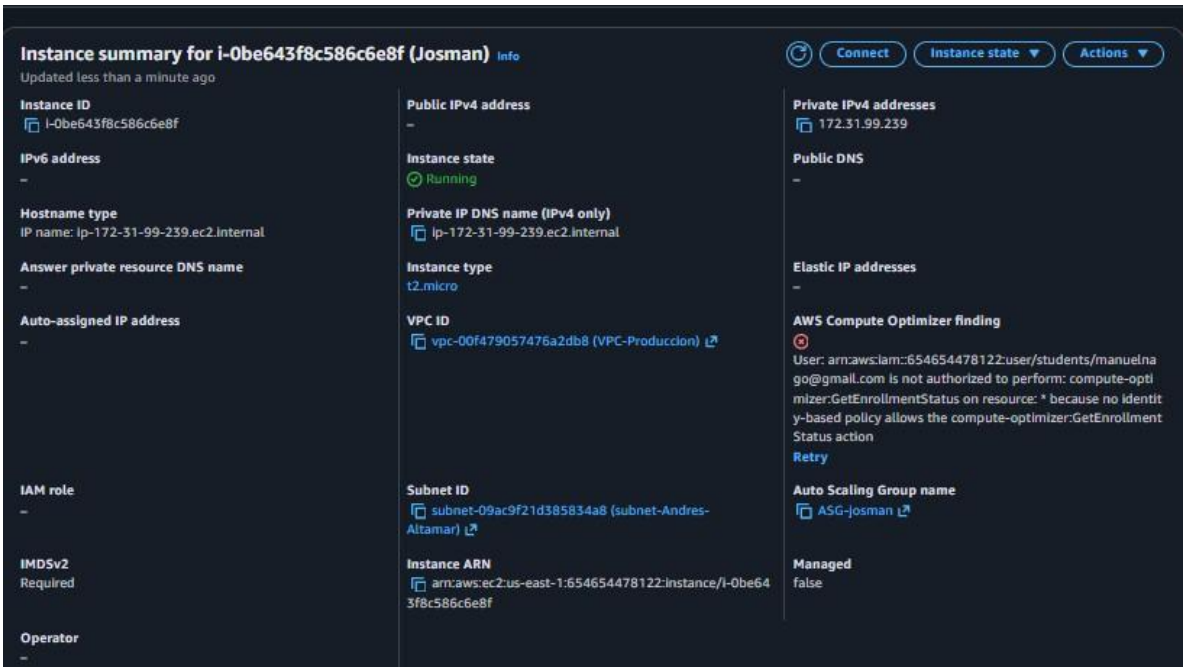
1. Launch Template: Crear una plantilla de lanzamiento (Launch Template) con una instancia t2.micro (Amazon Linux 2023).
2. Auto Scaling Group (ASG): * Crear un grupo que use el template anterior.

○ Configurar capacidad: Deseada: 1, Mínima: 1, Máxima: 2

Captura de pantalla de la consola de EC2 donde se vea el

Auto Scaling Group configurado y la instancia t2.micro en estado Running.





Parte 2: Crear una alarma en CloudWatch

1. Crea dos alarmas de CPU de la instancia que creó el auto scaling Group, una que sea sobre el 60% de CPU de advertencia y otra del 90% de CPU de alerta roja

3. Revisa desde la consola de cloudwatch todas las métricas asociadas a la instancia coloca por lo menos 3 donde se vea el id de la instancia.

Captura de pantalla de la sección de CloudWatch Alarms donde se vean las dos alarmas creadas (60% y 90%).

Alarms (2/30)

Clear selection | Create composite alarm | Actions | Create alarm

Filter alarms | Quick filters

Name	State	Actions	Actions muted - *new*	Last state update (UTC)	Conc
Alarmajosman-CPU-90	OK	No actions	Error	2026-03-17 01:39:16	CPUU minu
60% de CPU de advertencia	OK	No actions	Error	2026-03-17 01:38:19	CPUU minu
Alarmajosman-CPU-60	OK	No actions	Error	2026-03-17 01:37:25	CPUU minu
Alarma-Advertencia-60-Diana	OK	No actions	Error	2026-03-17 01:37:12	CPUU minu
CPU > 90 alerta roja i-0972eaf09b77c6433	Insufficient data	No actions	Error	2026-03-17 01:35:36	CPUU minu
pedro-alarma>90	Insufficient data	No actions	Error	2026-03-17 01:34:28	pedr 5 mi
Alarma CPU >60 i-0972eaf09b77c6433	OK	No actions	Error	2026-03-17 01:31:48	CPUU minu
Alarma crtica 90% CPU Julian Parra	OK	No actions	Error	2026-03-17 01:31:44	CPUU minu
Advertencia se Supera el 60% de CPU	Insufficient data	Actions enabled Warning	Error	2026-03-17 01:31:02	CPUU withi

Alarmajosman-CPU-60

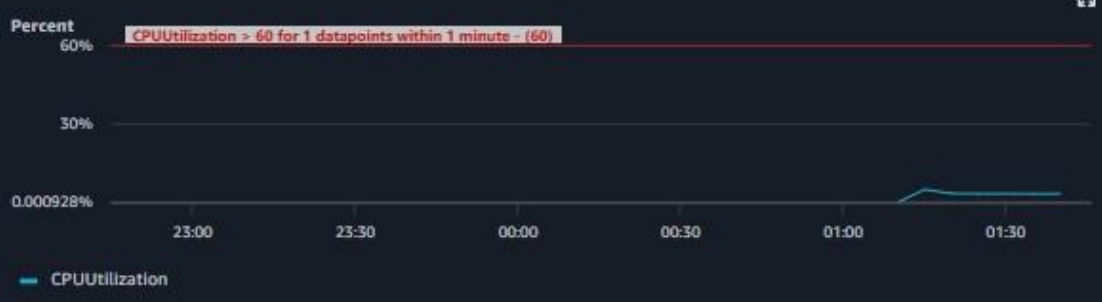
Actions | View | Explore related | Investigate

Graph

1h 3h 12h 1d 3d 1w Custom | UTC timezone

CPUUtilization OK

CPUUtilization > 60 for 1 datapoints within 1 minute



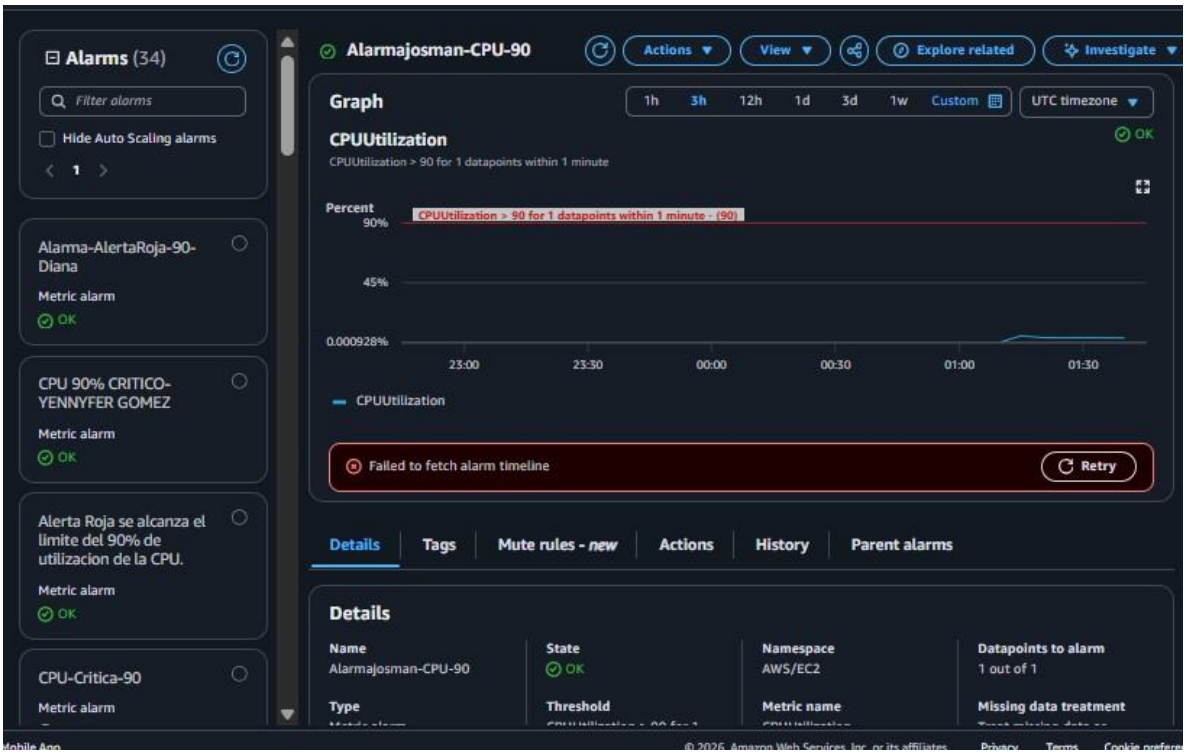
Failed to fetch alarm timeline | Retry

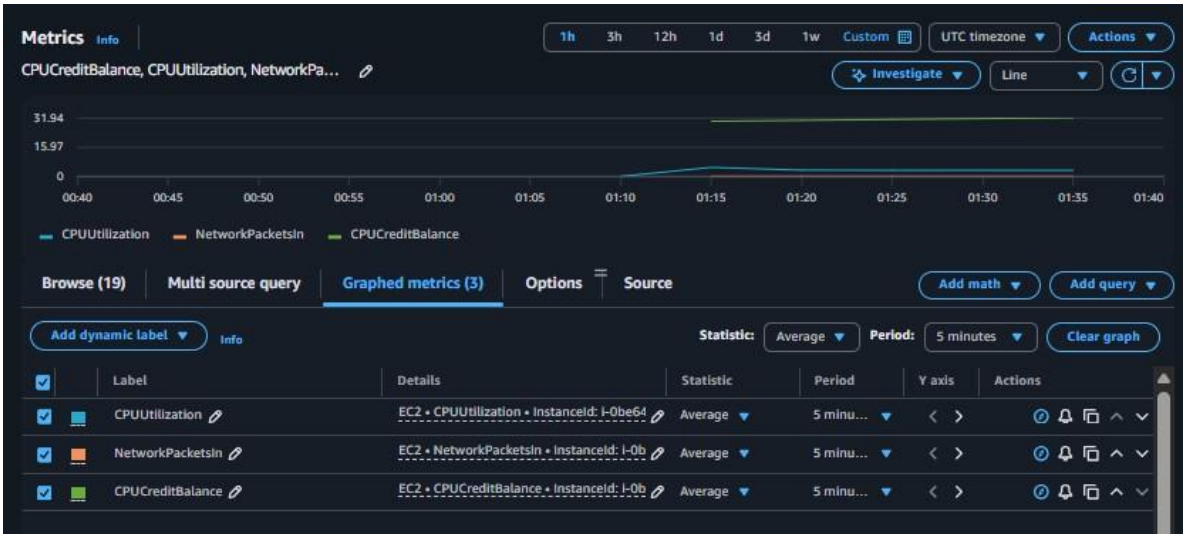
Details | Tags | Mute rules - *new* | Actions | History | Parent alarms

Details

Name	State	Namespace	Datapoints to alarm
Alarmajosman-CPU-60	OK	AWS/EC2	1 out of 1

Type	Threshold	Metric name	Missing data treatment





Parte 3: Revisar eventos en CloudTrail

1. En cloud Trail ingresa y busca el evento de creación de una de las instancias, pone el JSON e identifica desde donde se creó la instancia.

2. Revisa qué acciones ha realizado tu usuario de iam y pon las últimas 5 acciones, sin detalle.

El archivo JSON extraído de CloudTrail sobre la creación de la instancia (RunInstances).


```
{
  "eventVersion": "1.11",
  "userIdentity": {
    "type": "IAMUser",
    "principalId": "AIDAZQ3DSS4VJEMUHSOSM",
    "arn":
"arn:aws:iam::654654478122:user/students/manuelnago@gmail.com",
    "accountId": "654654478122",
    "accessKeyId": "ASIAZQ3DSS4VM725HJUY",
    "userName": "manuelnago@gmail.com",
    "sessionContext": {
      "attributes": {
        "creationDate": "2026-03-17T00:45:18Z",
        "mfaAuthenticated": "true"
      }
    },
    "invokedBy": "autoscaling.amazonaws.com"
```

```
},
"eventTime": "2026-03-17T01:13:52Z",
"eventSource": "ec2.amazonaws.com",
```

```
"eventName": "RunInstances",
"awsRegion": "us-east-1",
"sourceIPAddress": "autoscaling.amazonaws.com",
"userAgent": "autoscaling.amazonaws.com",
"errorCode": "Client.DryRunOperation",
"errorMessage": "Request would have succeeded, but DryRun flag is set.",
"requestParameters": {
  "instancesSet": {
    "items": [
      {
        "minCount": 1,
        "maxCount": 1
      }
    ]
  },
```

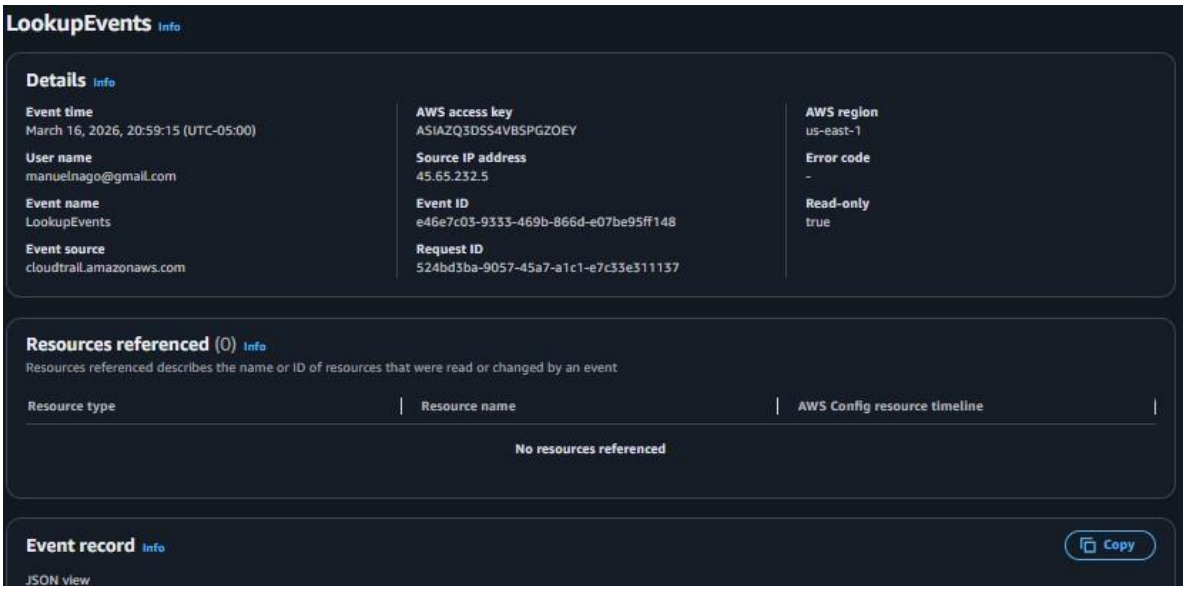
```
    "instanceType": "t2.micro",
    "blockDeviceMapping": {},
    "availabilityZone": "us-east-1a",
    "monitoring": {
        "enabled": false
    },
    "subnetId": "subnet-09ac9f21d385834a8",
    "disableApiTermination": false,
    "disableApiStop": false,
    "clientToken": "14f78969-31be-4468-abc0-3d04459b5f55",
    "launchTemplate": {
        "launchTemplateId": "lt-059b484c77530ce59",
        "version": "1"
    }
},
"responseElements": null,
"requestID": "7947a68b-cfed-4894-a8d3-b4881ddff77b",
"eventID": "6003f010-4ea3-4e59-8c0e-b4ff4f9c05ca",
"readOnly": false,
```

```
  "eventType": "AwsApiCall",

  "managementEvent": true,

  "recipientAccountId": "654654478122",

  "eventCategory": "Management",

  "sessionCredentialFromConsole": "true"

}
```


¿Desde qué dirección IP o servicio se originó la creación de la instancia según el JSON?

userAgent": "autoscaling.amazonaws.com sourceIPAddress": "autoscaling.amazonaws.com

5 eventos, solo 2 con JSON y otros con captura de pantalla




```
{
  "eventVersion": "1.11",
  "userIdentity": {
    "type": "IAMUser",
    "principalId": "AIDAZQ3DSS4VJEMUHSOSM",
    "arn": "arn:aws:iam::654654478122:user/students/manuelnago@gmail.com",
    "accountId": "654654478122",
    "accessKeyId": "ASIAZQ3DSS4VBSPGZOEY",
    "userName": "manuelnago@gmail.com",
    "sessionContext": {
      "attributes": {
        "creationDate": "2026-03-17T00:45:18Z",
        "mfaAuthenticated": "true"
      }
    }
  },
```

"eventTime": "2026-03-17T01:59:15Z",

"eventSource": "cloudtrail.amazonaws.com",

"eventName": "LookupEvents",

"awsRegion": "us-east-1",

"sourceIPAddress": "45.65.232.5",

"userAgent": "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/146.0.0.0 Safari/537.36",

"requestParameters": {

"maxResults": 50,

"nextToken": "Vts0wrPTK7o9iib7n5itltRhsruLmexb1rjLgTSzqbWQrfoP4BIe6znuyzjVYl2lD3zyzUneFIsaJvLuIRumtQ=="

},

"responseElements": null,

"requestID": "524bd3ba-9057-45a7-a1c1-e7c33e311137",

"eventID": "e46e7c03-9333-469b-866d-e07be95ff148",

"readOnly": true,

"eventType": "AwsApiCall",

"managementEvent": true,

"recipientAccountId": "654654478122",

"eventCategory": "Management",

"tlsDetails": {

"tlsVersion": "TLSv1.3",

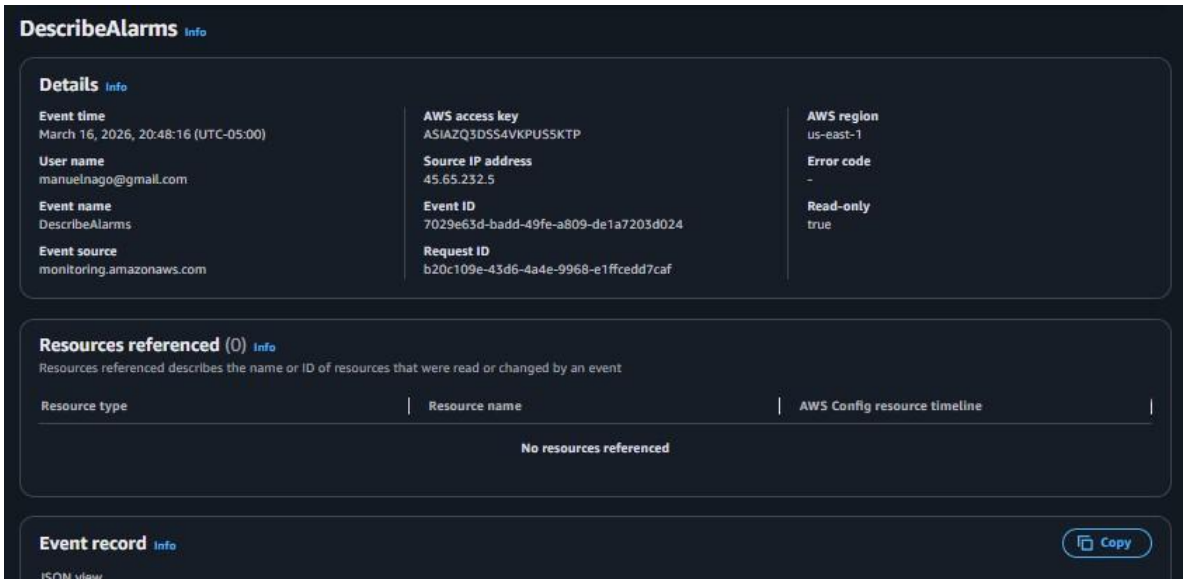
"cipherSuite": "TLS_AES_128_GCM_SHA256",

"clientProvidedHostHeader": "cloudtrail.us-east-1.amazonaws.com"

},

"sessionCredentialFromConsole": "true"

}



{

"eventVersion": "1.11",

"userIdentity": {

"type": "IAMUser",

"principalId": "AIDAZQ3DSS4VJEMUHSOSM",

"arn": "arn:aws:iam::654654478122:user/students/manuelnago@gmail.com",

"accountId": "654654478122",

"accessKeyId": "ASIAZQ3DSS4VKPUS5KTP",

"userName": "manuelnago@gmail.com",

"sessionContext": {

"attributes": {

"creationDate": "2026-03-17T00:45:18Z",

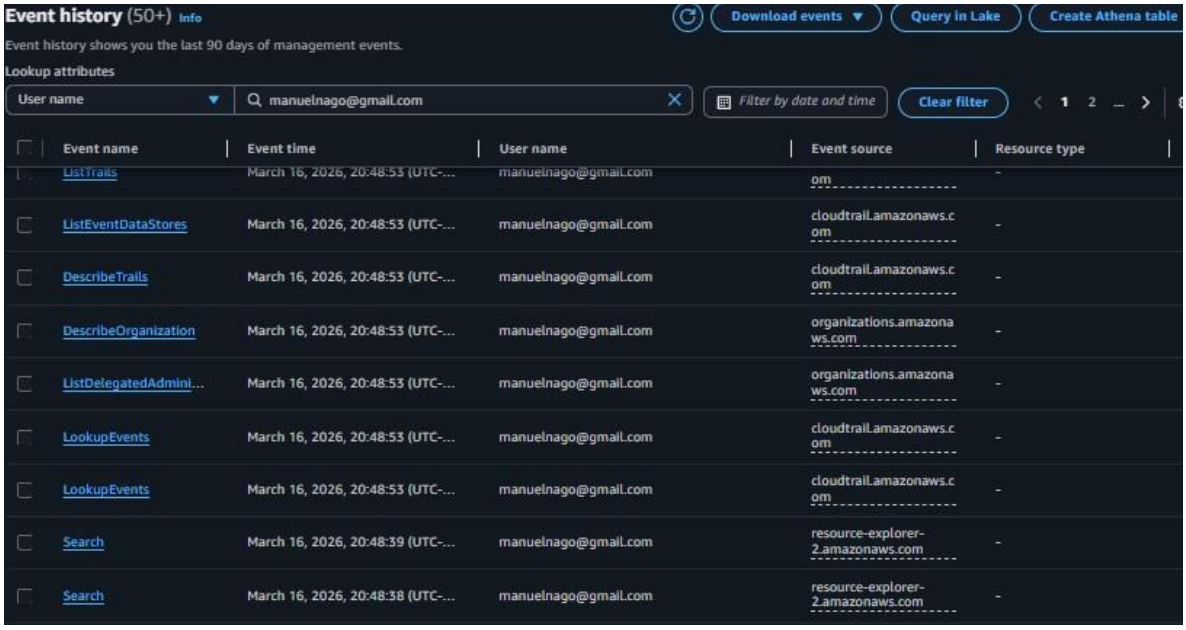
"mfaAuthenticated": "true"

}

}

```
  },
  "eventTime": "2026-03-17T01:48:16Z",
  "eventSource": "monitoring.amazonaws.com",
  "eventName": "DescribeAlarms",
  "awsRegion": "us-east-1",
  "sourceIPAddress": "45.65.232.5",
  "userAgent": "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36
(KHTML, like Gecko) Chrome/146.0.0.0 Safari/537.36",
  "requestParameters": {
    "alarmNames": [
      "Alarmajosman-CPU-60"
    ],
    "alarmTypes": [
      "MetricAlarm"
    ],
    "excludeAutoscalingAlarms": false,
    "maxRecords": 100
  },
  "responseElements": null,
  "requestID": "b20c109e-43d6-4a4e-9968-e1ffcedd7caf",
  "eventID": "7029e63d-badd-49fe-a809-de1a7203d024",
  "readOnly": true,
  "eventType": "AwsApiCall",
  "managementEvent": true,
  "recipientAccountId": "654654478122",
  "eventCategory": "Management",
```

```
  "tlsDetails": {
    "tlsVersion": "TLSv1.3",
    "cipherSuite": "TLS_AES_128_GCM_SHA256",
    "clientProvidedHostHeader": "monitoring.us-east-1.amazonaws.com"
  },
  "sessionCredentialFromConsole": "true"
}
```

Parte 4 : Creación de función lambda

1. En esta actividad deberá crear una función AWS Lambda que consulte las instancias EC2

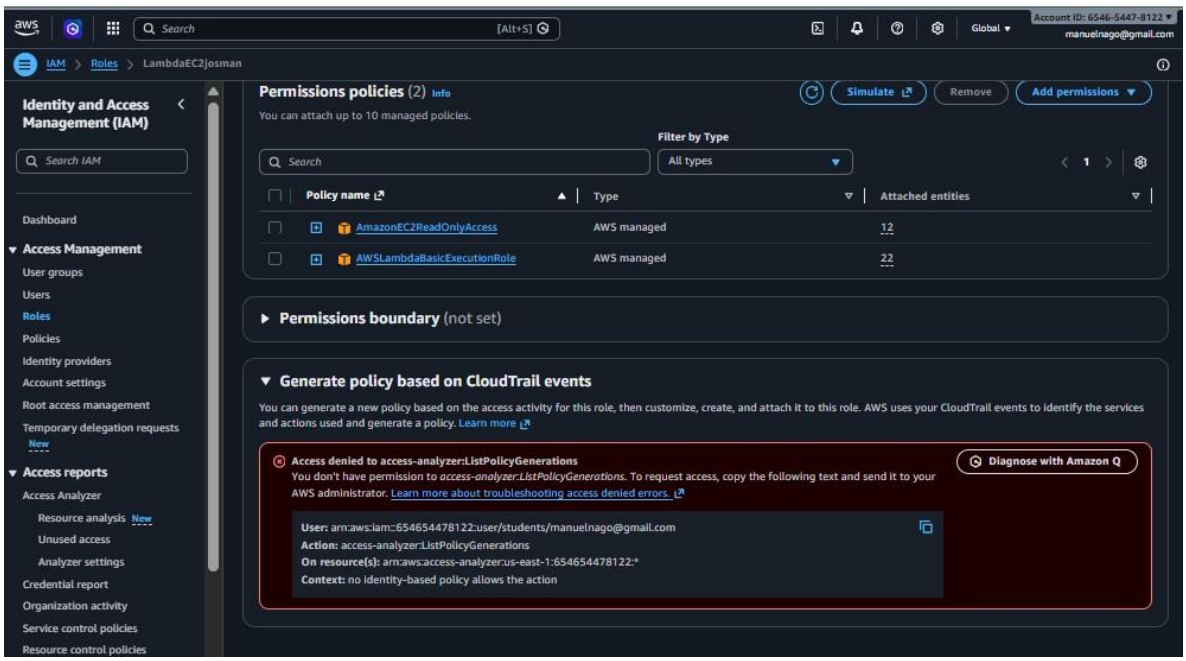
2. Muestra la siguiente información de cada una.

- ID de la instancia
- Estado
- Tipo de instancia

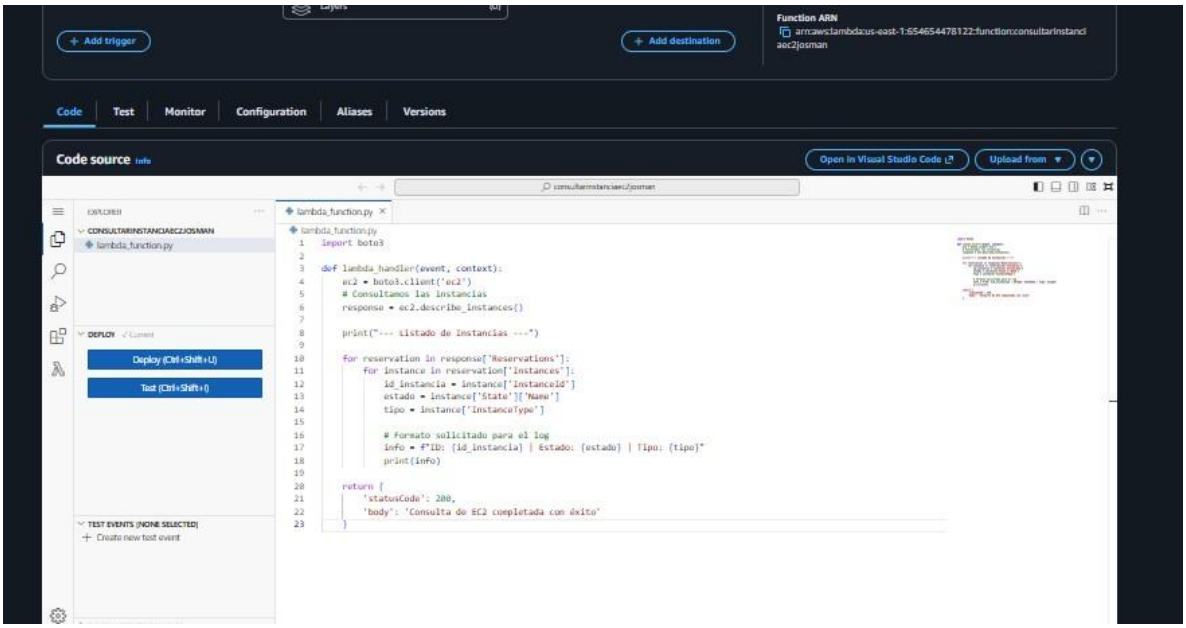
3. Crear un IAM Role para la función Lambda

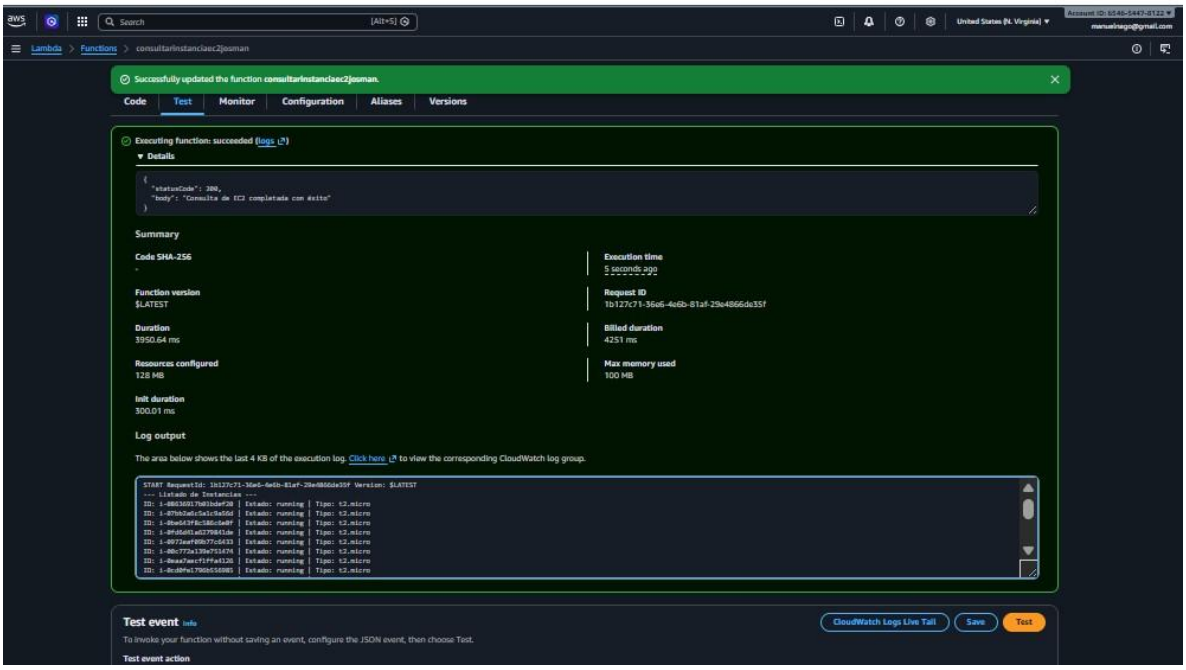
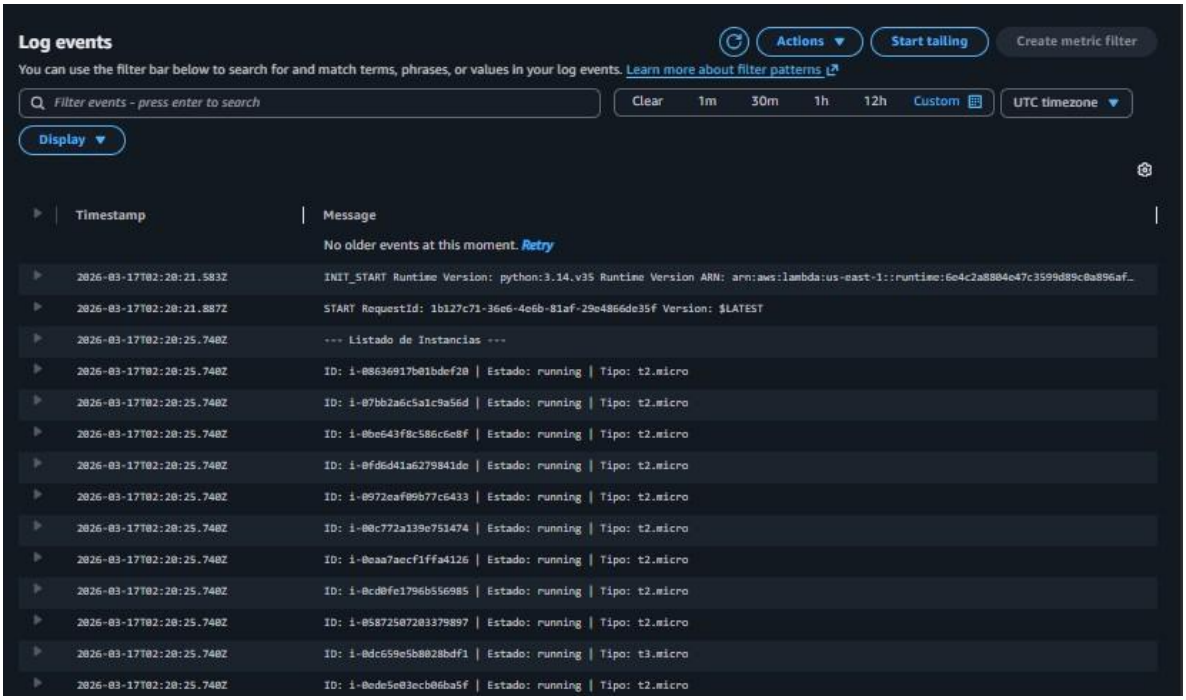
4. Verificar en CloudWatch Logs que la función haya registrado correctamente la información de las instancias, puntualmente el log de la lambda que implementó.

- **El rol de la lambda**



- **El código de la lambda**


+ Add trigger
+ Add destination
Function ARN
arn:aws:lambda:us-east-1:654654478122:function:consultarInstanciaec2josman
Code
Test
Monitor
Configuration
Aliases
Versions
Code source Info
Open in Visual Studio Code
Upload from
EXPLORER
CONSULTARINSTANCIAEC2JOSMAN
lambda_function.py
DEPLOY
Deploy (Ctrl+Shift+U)
Test (Ctrl+Shift+I)
TEST EVENTS (NONE SELECTED)
Create new test event
import boto3
def lambda_handler(event, context):
ec2 = boto3.client('ec2')
# Consultamos las instancias
response = ec2.describe_instances()
print("--- Listado de Instancias ---")
for reservation in response['Reservations']:
for instance in reservation['Instances']:
id_instancia = instance['InstanceId']
estado = instance['State']['Name']
tipo = instance['InstanceType']
# Formato solicitado para el log
info = f"ID: {id_instancia} | Estado: {estado} | Tipo: {tipo}"
print(info)
return {
'statusCode': 200,
'body': 'Consulta de EC2 completada con éxito'
}


Lambda > Functions > consultarInstanciaec2josman
Successfully updated the function consultarInstanciaec2josman.
Code
Test
Monitor
Configuration
Aliases
Versions
Executing function: succeeded (logs)
Details
{
"statusCode": 200,
"body": "Consulta de EC2 completada con éxito"
}
Summary
Code SHA-256
Execution time
5 seconds ago
Function version
$LATEST
Request ID
1b127c71-36e6-4e6b-81af-29e4866de35f
Duration
3950.64 ms
Billed duration
4251 ms
Resources configured
128 MB
Max memory used
100 MB
Init duration
300.01 ms
Log output
The area below shows the last 4 KB of the execution log. Click here to view the corresponding CloudWatch log group.
START RequestId: 1b127c71-36e6-4e6b-81af-29e4866de35f Version: $LATEST
--- Listado de Instancias ---
ID: i-0863691fb01bdef20 | Estado: running | Tipo: t2.micro
ID: i-07bb2a6c5e1c9a66d | Estado: running | Tipo: t2.micro
ID: i-0ba443f8c586c6a8f | Estado: running | Tipo: t2.micro
ID: i-0fd6d41a6279841de | Estado: running | Tipo: t2.micro
ID: i-0973eaf08b77c6433 | Estado: running | Tipo: t2.micro
ID: i-0bc772a139a753474 | Estado: running | Tipo: t2.micro
ID: i-0eae7aecf1ff4a126 | Estado: running | Tipo: t2.micro
ID: i-0cd0fw17906558085 | Estado: running | Tipo: t2.micro
Test event Info
CloudWatch Logs Live Tail
Save
Test
To invoke your function without saving an event, configure the JSON event, then choose Test.
Test event action

○ **El log group de la lambda donde muestra el mensaje que esta deja en la ejecución la lambda.**